

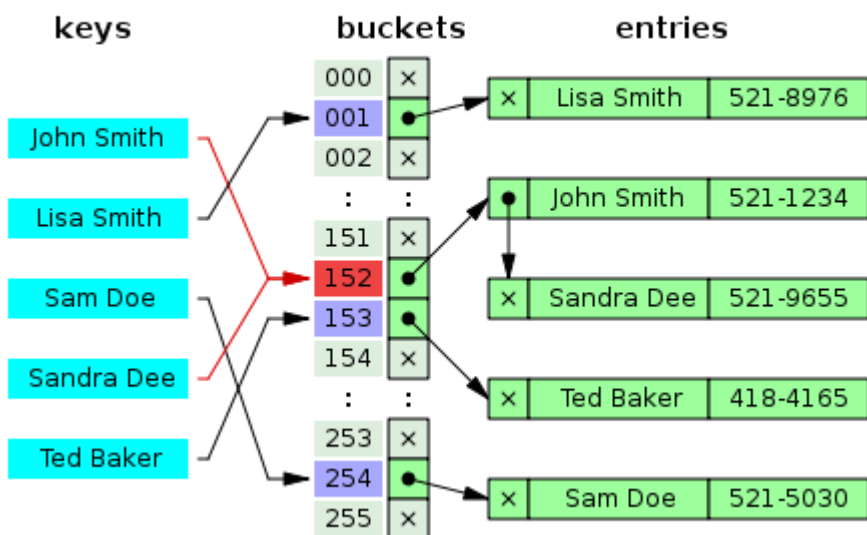
Consistent Hashing

Hash Tables

- Goal: Efficiently store and search data
 - Naive approach: Array or linked list
 - Search by iterating elements → **$O(n)$** complexity
 - Fast for small lists, slow for large datasets
 - Hash Functions:
 - Map arbitrary data → integer index
 - Allows near-constant time access for non-sequential or random keys
 - Formula: $\text{index} = \text{hash}(\text{object}) \bmod N$
 - N = array size (practical, usually $\sim 2 \times$ expected elements)
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Handling Collisions

- Multiple objects can map to the same index → **collisions**
- Typical solution: **buckets**
 - Each array entry points to a list of objects
 - Insert: add to bucket if not present
 - Search: check bucket for object
- Complexity: $\sim O(N/k)$, with k = number of buckets



Distributed Hash Tables & Key Distribution

- **Motivation:**
 - Split a hash table across multiple servers to bypass memory limits
 - Enable arbitrarily large hash tables
- **How It Works:**
 - Objects and keys are distributed among several servers → **Distributed Hash Table (DHT)**

- Typical use case: **in-memory caches** (e.g., **Memcached**)

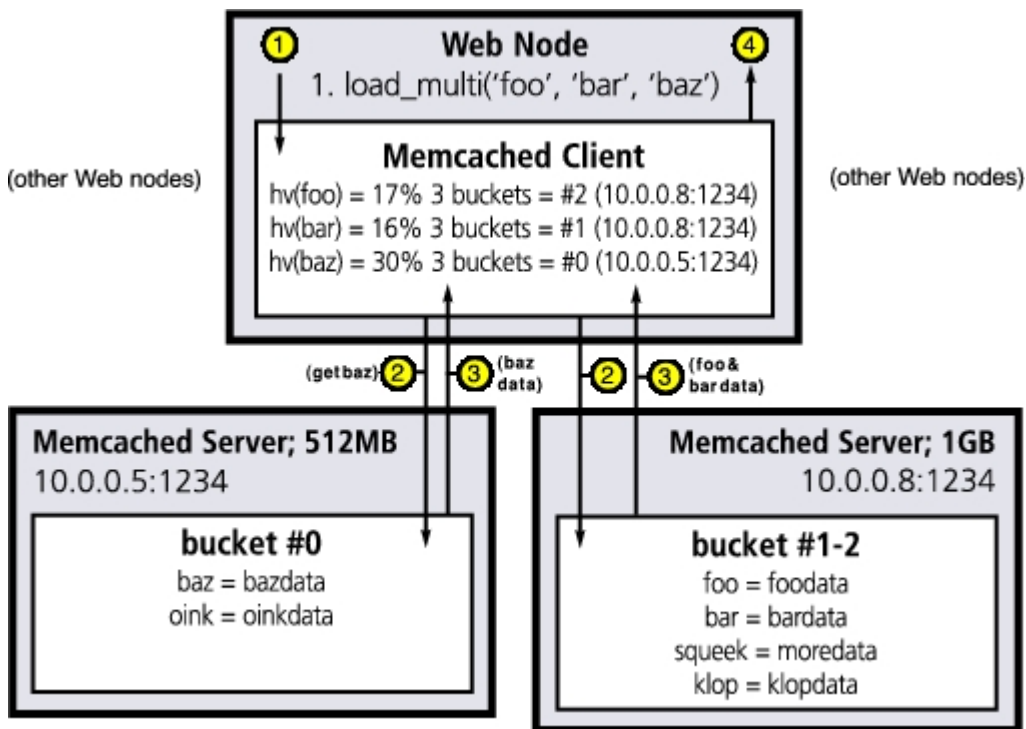
- Pool of servers stores key/value pairs
- Provides fast access to frequently used data

Cache Workflow

- Application queries the cache first
- If **cache miss**, fetch from database/source
- Store result in cache for future access

- Key Distribution:**

- Strategy determines **which keys go to which servers**
- Crucial for **load balancing, scalability, and reliability**



Key Distribution via Hash Modulo

- Formula:

```
server_index = hash(key) mod N
```

- N = number of servers
- Client computes hash → modulo → contacts server
- Same hash function across clients is required

Example Setup

- Servers: **A, B, C** (N = 3)
- Keys and their hashes:

KEY	HASH	HASH mod 3
john	1633428562	2
bill	7594634739	0

jane	5000799124	1
steve	9787173343	0
kate	3421657995	2

A	B	C
bill	jane	john
steve		kate

The Rehashing Problem

- Simple modulo-based distribution works **until the number of servers changes**
- Problem occurs when:
 - A server crashes or becomes unavailable
 - New servers are added to the pool
- Most keys need to be **rehashed** because `hash(key) mod N` changes

Example: Removing Server C

- Previous pool: 3 servers (A, B, C)
- Remove server C → N = 2
- Recompute server index: `server_index = hash(key) mod 2`

KEY	HASH	HASH mod 2
john	1633428562	0
bill	7594634739	1
jane	5000799124	0
steve	9787173343	1
kate	3421657995	1

A	B
john	bill
jane	steve
	kate

- Observation:** All key locations changed, not just those on removed server

Consequences for Caching

- Most queries result in **cache misses**
- Data must be fetched again from the source
- Heavy load on origin servers (e.g., databases)
- Can severely **degrade performance** or cause **server crashes**

The Solution: Consistent Hashing

- Problem: Modulo-based distribution requires massive rehashing when servers change

- Solution: **Consistent Hashing** (Karger et al., 1997)
 - Minimizes key relocation when servers are added or removed
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Hash Ring Concept

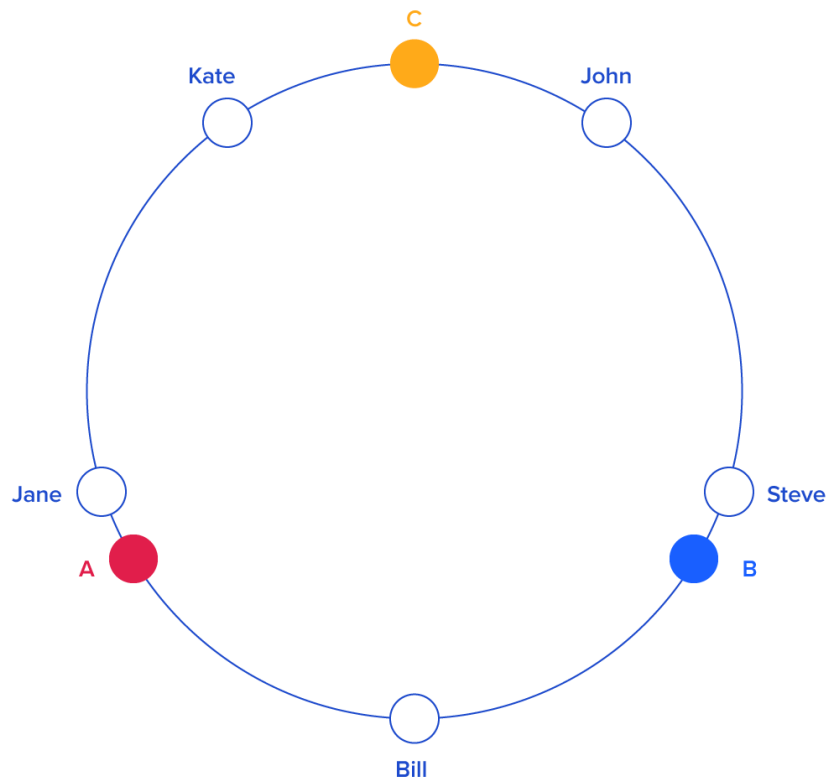
- Map the hash output range to a circle:
 - $0 \rightarrow 0^\circ$
 - `INT_MAX` $\rightarrow 360^\circ$
 - All other hashes $\rightarrow$ intermediate angles
 - Keys and servers are placed on the same circle by hashing their values
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Example: Keys on the Circle

KEY	HASH	ANGLE (DEG)
john	1633428562	58.8
bill	7594634739	273.4
jane	5000799124	180
steve	9787173343	352.3
kate	3421657995	123.2

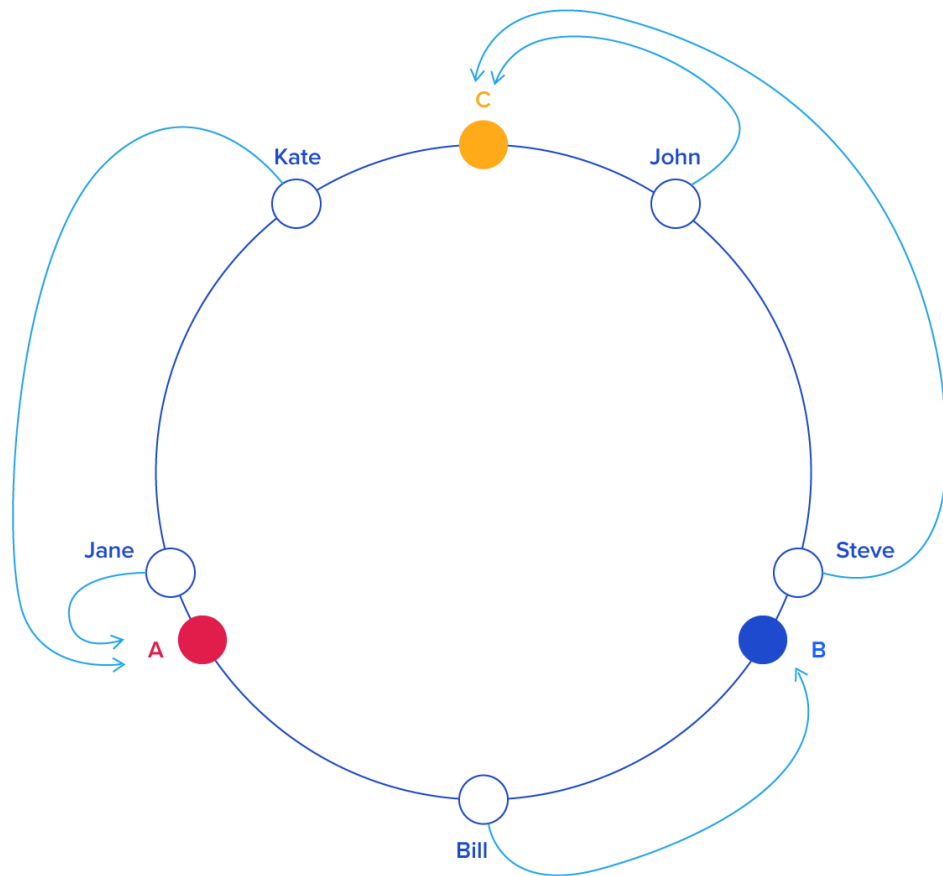
- Servers are also hashed to angles:

SERVER	HASH	ANGLE (DEG)
A	5572014558	200.6
B	8077113362	290.8
C	2269549488	81.7



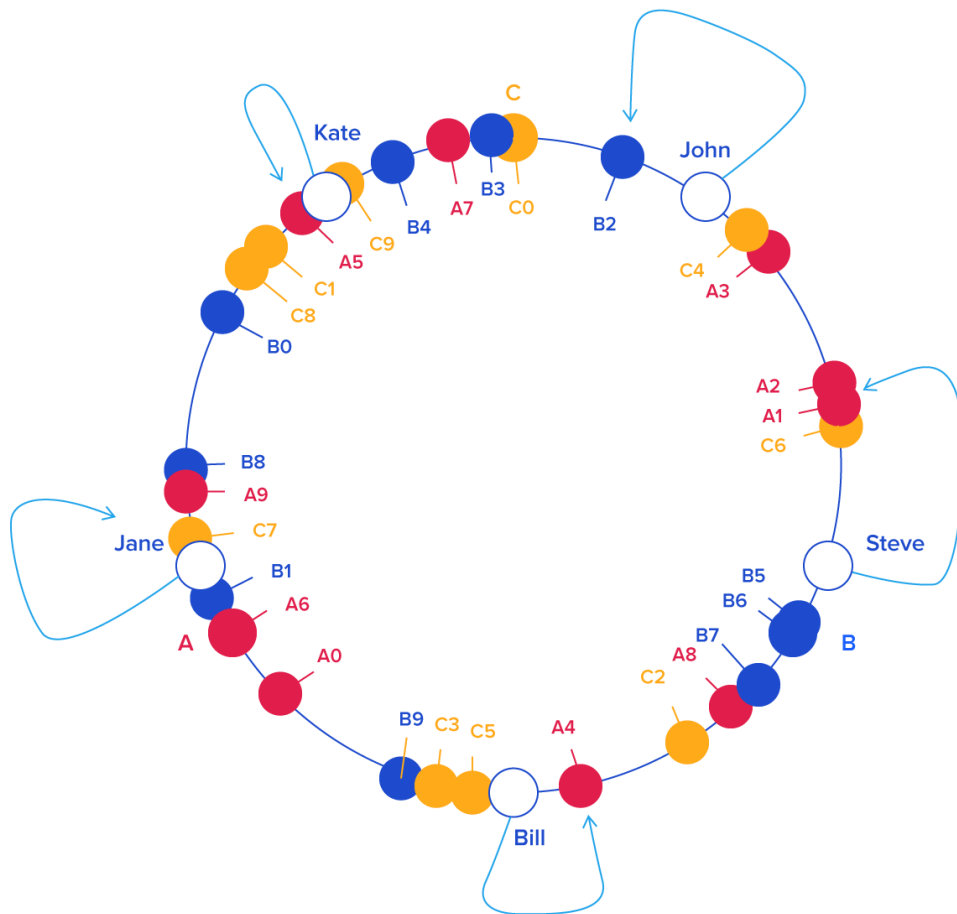
Mapping Keys to Servers

- Rule: **Each key goes to the first server clockwise after its angle**
- Example assignments:



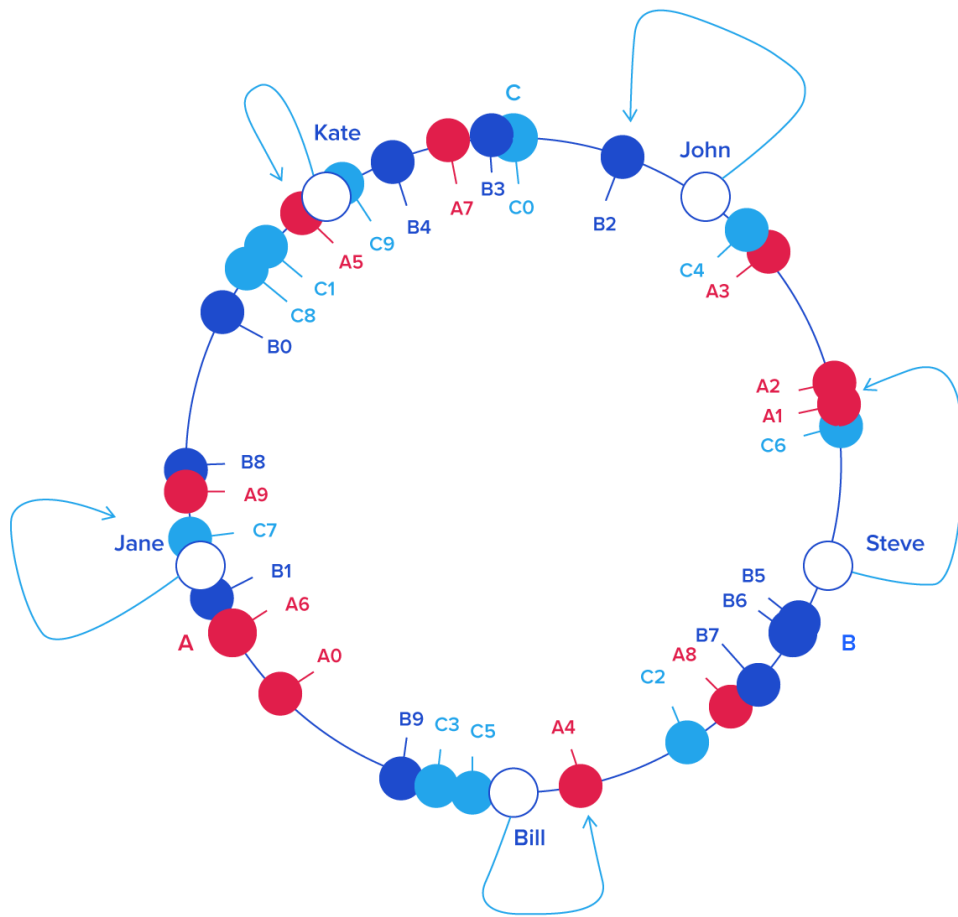
Balancing Load: Virtual Nodes

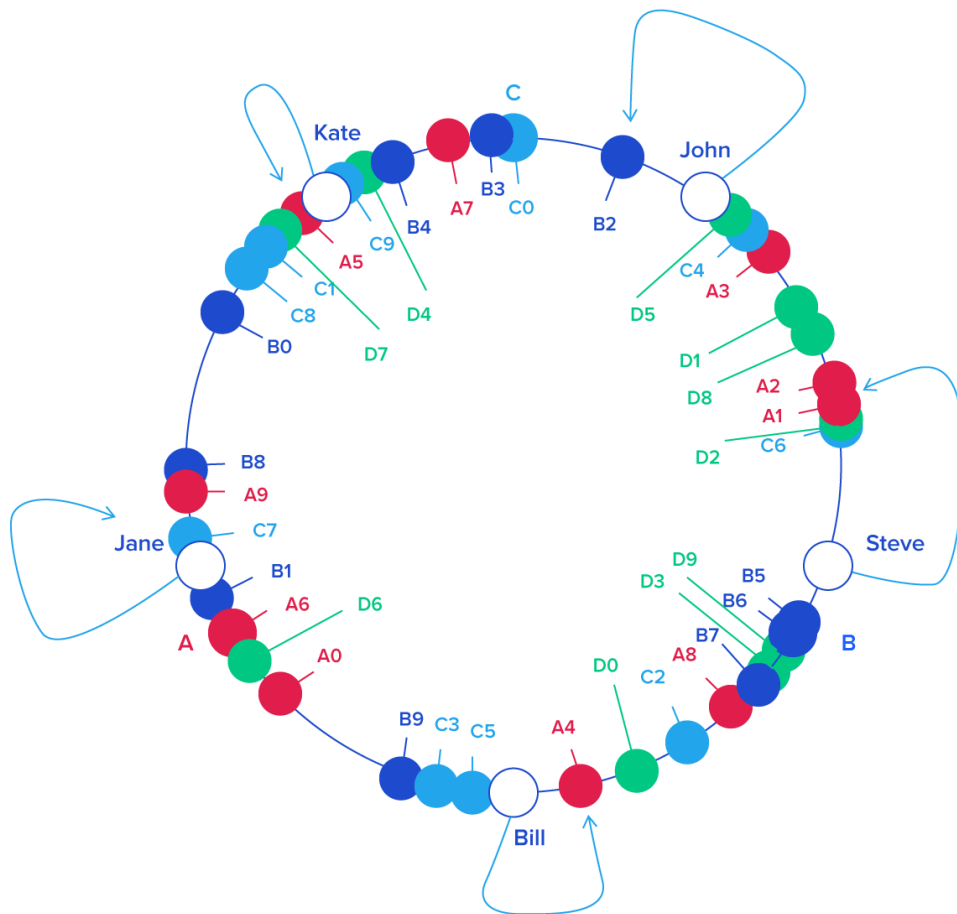
- Assign multiple **labels/angles per server** (virtual nodes)
- Example: A0..A9, B0..B9, C0..C9
- Adjust weight per server for load balancing:
 - More powerful servers → more virtual nodes



Resilience to Server Changes

- **Server removal:** only keys mapped to removed server labels are reassigned
- **Server addition:** only keys near new server labels are reassigned
- Keys for other servers **remain unchanged**
- Result: **minimal remapping** → avoids massive cache misses





Replication Factor

- Store each key on **k different servers**
- Ensures **fault tolerance**: data available if a server fails
- Example: Replication factor = 3 → key `john` on B, C, and A

Resources

- <https://www.linuxjournal.com/article/7451>
- <https://www.toptal.com/developers/big-data/consistent-hashing>